HU Collection

Engineering and Science Admissions Test

Mock Set F

Mathematics 2

Time allowed: 40 minutes
Questions: 27 multiple choice
Calculators: not permitted
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Date written: 30 October 2026

Name: Date sat: Score: / 27

Instructions

Answer all 27 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers, so answer every question. Calculators are not permitted. This paper is deliberately longer than the time allows: move on from any question that is costing you more than two minutes, and come back to it at the end.

Q	1	2	3	4	5	6	7	8	9	10
Ans										

Q	11	12	13	14	15	16	17	18	19	20
Ans										

Q	21	22	23	24	25	26	27			
Ans										

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1. A metal component is heated in a furnace for two hours. Its temperature, in °C, t hours after heating begins is modelled by

$$\theta = 20 + 27t - kt^2 + t^3, \qquad 0 \le t \le 2,$$

where k is a constant set on the furnace controller. To avoid cracking, two rules must hold throughout the two hours: the temperature must never decrease, and it must never rise at a rate greater than 27 °C per hour.

Which of the following describes all the values of k for which both rules hold?

A $3 \le k \le 15$ **B** $3 \le k \le \dfrac{39}{4}$ **C** $-9 \le k \le 9$

D $\dfrac{3}{2} \le k \le \dfrac{39}{4}$ **E** $0 \le k \le \dfrac{39}{4}$ **F** $3 \le k \le 9$

2. A sensor's calibration routine accepts a whole-number reading x with $4 \le x \le 100$ and computes the output

$$f(x) = \sqrt{x + 4\sqrt{x-4}} + \sqrt{x - 4\sqrt{x-4}}.$$

A reading is *accepted* if the output $f(x)$ is an integer. How many of the 97 possible readings are accepted?

A 10 **B** 97 **C** 13

D 12 **E** 8 **F** 5

3. A cyclist riding along a straight, level road applies her brakes at time $t = 0$ and squeezes them progressively harder, so that while she is braking her deceleration at time t seconds is $k\sqrt{t}$ m s^{-2}, where k is a positive constant. She comes to rest exactly 4 seconds after she starts to brake, having travelled 24 m while braking.

What was her speed, in m s^{-1}, at the moment she started to brake?

A 10 **B** 6 **C** 8

D 12 **E** 15 **F** 9

4. A designer draws a spiral pattern around a point O by shading infinitely many triangles, each of which is an isosceles right-angled triangle. The first triangle is OA_0A_1, with hypotenuse OA_0 of length 16 cm and its right angle at A_1. For each $n \ge 1$, the next triangle OA_nA_{n+1} has hypotenuse OA_n (one of the two shorter sides of the previous triangle) and its right angle at A_{n+1}, and it is drawn on the opposite side of the line OA_n from the previous triangle, so that the pattern winds anticlockwise around O. What is the area, in cm^2, of the shaded region? (A point that is shaded more than once is counted only once.)

A $\dfrac{255}{4}$ **B** 120 **C** $\dfrac{256}{3}$

D $\dfrac{255}{2}$ **E** 127 **F** 128

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5. A field lies between a straight hedge along the x-axis and a stream whose course is the curve $y = \sqrt{x}$, and it is closed off at the ends by the lines $x = 0$ and $x = 9$, with lengths in units of 10 m. A surveyor estimates the area of the field using two trapezia. She measures the width y of the field at the two ends and at one further point $x = c$, where $0 < c < 9$ and she may choose c freely. She then applies the trapezium rule to the two strips $0 \le x \le c$ and $c \le x \le 9$, which need not have equal widths.

She chooses c to make her estimate as accurate as possible. By how much, in square units, does this best estimate differ from the true area of the field?

A $\dfrac{9}{8}$ **B** $\dfrac{9}{2}$ **C** $\dfrac{9}{4}$

D $\dfrac{3}{4}$ **E** $\dfrac{3}{2}$ **F** $\dfrac{45 - 27\sqrt{2}}{4}$

6. Seismologists model the energy E joules released by an earthquake of magnitude M by

$$\log_{10} E = 1.5M + 4.8.$$

A main shock of magnitude 7.0 is followed by a sequence of aftershocks. In an idealised model there are infinitely many aftershocks: the first has magnitude 6.2, and each later aftershock has magnitude 0.1 less than the one before it. In this question take $\log_{10} 2$ to be exactly 0.3. The total energy released by all the aftershocks together is what fraction of the energy released by the main shock?

A $\dfrac{1+\sqrt{2}}{16}$ **B** $\dfrac{1}{16}$ **C** $\dfrac{1}{8}$

D $\dfrac{2+\sqrt{2}}{16}$ **E** $\dfrac{2+\sqrt{2}}{8}$ **F** $\dfrac{\sqrt{2}}{8}$

7. An engineer is choosing the internal radius r of a long water main. Over the working life of the main, the cost of the pipe itself is proportional to $r^{3/2}$, and the cost of the energy used to pump water through it at the required rate is inversely proportional to r^4. The radius is always chosen to make the sum of these two costs as small as possible.

The required flow rate is then doubled. For every radius this makes the pumping cost 4 times as large as before, while the cost of the pipe is unchanged, and the radius is chosen again in the same way. By what factor does the least possible total cost increase?

A 2 **B** $2^{16/11}$ **C** $2^{6/11}$

D $2^{3/4}$ **E** 4 **F** $2^{4/11}$

8. A slow-responding thermometer is placed in a room at time $t = 0$, where t is measured in hours. For $t \ge 1$ the reading on the thermometer at time t is the mean value of the actual room temperature over the previous hour, that is,

$$\text{reading at time } t = \int_{t-1}^{t} T(s)\, ds,$$

where $T(s)$ °C is the actual temperature at time s. The actual temperature is known to be a quadratic function of time, and for $1 \le t \le 6$ the readings are $t^2 - 6t + 12$ °C.

What was the lowest actual temperature in the room, in °C, during the period $0 \le t \le 6$?

A $\dfrac{11}{4}$ **B** $\dfrac{37}{12}$ **C** $\dfrac{19}{6}$

D 3 **E** $\dfrac{17}{6}$ **F** $\dfrac{35}{12}$

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9. Three radio transmitters stand at the points $A(0,0)$, $B(9,0)$ and $C(6,9)$, with distances in kilometres. A survey drone flies round and round the circle

$$x^2 + y^2 - 4x - 6y - 12 = 0.$$

When the drone is at the point P, its *signal load* is defined to be $PA^2 + PB^2 + PC^2$. What is the greatest signal load, in km^2, that the drone experiences?

A 324 **B** 288 **C** 108

D 264 **E** 180 **F** 192

10. A garden hose nozzle is at ground level at the origin, and the jet of water it produces travels in a vertical plane along the curve

$$y = kx - \frac{\left(1 + k^2\right) x^2}{20},$$

where x and y are the horizontal and vertical distances in metres and k is the tangent of the angle of elevation of the nozzle. The mounting only allows angles with $0 < k \leq 2$. A vertical wall stands on level ground at horizontal distance X metres from the nozzle, and the jet is aimed straight at it. For which values of X is it possible, for a suitable choice of k, for the jet to hit the wall at a point at least 3 metres above the ground?

A $0 < X \leq 2\sqrt{10}$ **B** $\sqrt{10} \leq X \leq 2\sqrt{10}$ **C** $5 \leq X \leq 2\sqrt{10}$

D $2 \leq X \leq 5$ **E** $2 \leq X \leq 6$ **F** $2 \leq X \leq 2\sqrt{10}$

11. A surveyor walks along a straight, level road and measures the angle of elevation of the top of a vertical radio mast. The foot of the mast is on the same level ground, but not on the road. At three points A, B and C of the road, in that order, with $AB = 60$ m and $BC = 20$ m, the tangents of the angles of elevation are $\frac{2}{3}$, $\frac{4}{5}$ and $\frac{3}{4}$ respectively. What is the height of the mast?

A 72 m **B** 60 m **C** 50 m

D 45 m **E** 48 m **F** 40 m

12. The cross-section of an arch is the parabola $y = x^2$. A circular hoop, lying in the same plane, touches the parabola at the point $P(1,1)$, so that the circle and the parabola have the same tangent there, and also passes through the point $Q(2,4)$ of the parabola. The hoop meets the parabola at exactly one other point R. What is the area of the triangle PQR?

A 21 **B** 10 **C** 3

D 30 **E** 15 **F** 6

13. A designer's template has a curved edge in the shape of the curve

$$y = x^4 - 2x^3 - 12x^2 + 5x + 1.$$

There are exactly two points on this curve at which the gradient $\dfrac{dy}{dx}$, regarded as a function of x, has a stationary value; call them I and J, where I is to the left of J. A straight cut is made along the line through I and J. This line meets the curve at exactly two further points, P and Q, and the four points lie along the line in the order P, I, J, Q.

What is the ratio $PI : IJ$?

A $(\sqrt{5} - 2) : 1$ **B** $(\sqrt{5} + 1) : 2$ **C** $1 : \sqrt{5}$

D $(\sqrt{5} - 1) : 2$ **E** $1 : 1$ **F** $1 : 2$

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14. A circular pond has boundary $x^2 + y^2 = 25$, with distances in metres. A gardener walks from $A(-10, 0)$ to $B(5, 5)$. She may not enter the pond, but she may walk along its edge. What is the length, in metres, of the shortest route she can take?

A $10 + \dfrac{5\pi}{6}$ **B** $5\sqrt{10}$ **C** $5 + 5\sqrt{3} + \dfrac{5\pi}{3}$

D $5\sqrt{2} + 5\sqrt{3} + \dfrac{5\pi}{6}$ **E** $5 + 5\sqrt{3} + \dfrac{5\pi}{6}$ **F** $5 + 5\sqrt{3} + \dfrac{10\pi}{3}$

15. A designer is building a frame along a curved rail whose profile is the curve $y = x^3 - kx$, where k is a real constant. A straight brace is fixed along the tangent to the rail at a point P other than the origin, and this tangent meets the rail again at a second point Q. A second straight brace is fixed along the tangent to the rail at Q. For stability the two braces must be perpendicular to each other.

Which of the following describes all the values of k for which at least one position of P makes this possible?

A $k \geq \dfrac{4}{3}$ **B** $k \geq \dfrac{2}{3}$ **C** $k \geq \dfrac{16}{9}$

D $k \geq 1$ **E** $k \leq -\dfrac{4}{3}$ or $k \geq \dfrac{4}{3}$ **F** $k > 0$

16. What is the value of

$$\sum_{k=0}^{10} (-1)^k \binom{12}{k}\binom{12}{10-k} = \binom{12}{0}\binom{12}{10} - \binom{12}{1}\binom{12}{9} + \binom{12}{2}\binom{12}{8} - \cdots + \binom{12}{10}\binom{12}{0}\,?$$

A 495

B -792

C -495

D 0

E 924

F -924

G 792

17. A ticket office opens at time $t = 0$, where t is measured in hours, and customers stop arriving at $t = 10$. Model the customers as a continuous flow: they arrive at a rate of

$$r(t) = 30t - 3t^2 \text{ customers per hour}, \qquad 0 \leq t \leq 10,$$

and the office can serve at most 63 customers per hour. Whenever customers arrive faster than they can be served, the excess join a queue. While there is a queue the office serves at its full rate of 63 per hour, and there is no queue at $t = 0$. The total time spent waiting in the queue by all the customers, in customer-hours, is equal to the area under the graph of the number of customers in the queue against time.

What is the total time spent waiting in the queue by all the customers, in customer-hours?

A 96 **B** 64 **C** 144

D 32 **E** 108 **F** 72

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18. The profile of a test track is the curve $y = x^3 - 3x$, with lengths in metres. A horizontal laser beam, along the line $y = k$, is adjusted so that it meets the profile at exactly three distinct points. Let D be the distance between the two outermost of these three points.

As k varies over all the values for which there are three distinct meeting points, which of the following is the set of all possible values of D?

A $2\sqrt{3} \le D < 4$ **B** $0 < D \le 2\sqrt{3}$ **C** $2 < D \le 2\sqrt{3}$

D $3 < D < 4$ **E** $3 \le D \le 2\sqrt{3}$ **F** $3 < D \le 2\sqrt{3}$

19. A conventional car windscreen wiper has a straight arm pivoted at a fixed point O, and its blade lies along the arm, occupying the part of the arm from 30 cm to 70 cm from O. In one sweep the arm turns about O through an angle of $\frac{5\pi}{6}$ radians.

A bus has a pantograph wiper instead. Its rigid arm OA has length 50 cm and is pivoted at O, and the blade is a straight segment of length 40 cm whose midpoint is attached at A. A linkage keeps the blade vertical at all times. In one sweep the arm turns about O anticlockwise through an angle of $\frac{5\pi}{6}$ radians, starting from the horizontal position with A to the right of O.

Both blades are 40 cm long, and in both cases the midpoint of the blade moves along a circular arc of radius 50 cm through the same angle. By how many cm^2 does the area wiped in one sweep by the conventional wiper exceed the area wiped in one sweep by the pantograph wiper?

A $\dfrac{5000\pi}{3} - 2000\sqrt{3}$ **B** 0 **C** $\dfrac{5000\pi}{3} - 2000 - 1000\sqrt{3}$

D $\dfrac{5000\pi}{3} - 1000\left(\sqrt{6} + \sqrt{2}\right)$ **E** $\dfrac{2500\pi}{3} - 1000\sqrt{3}$ **F** $\dfrac{5000\pi}{3} - 3000$

20. A floodlight is fixed at the top of a vertical pole of height 6.75 m, whose foot is at the origin O. To the right of the pole the ground has the profile

$$y = x^3 - 6x^2 + 9x, \qquad 0 \le x \le 4,$$

where x m is the horizontal distance from O and y m is the height of the ground: a hill rises to its summit at $x = 1$, falls to a hollow at $x = 3$ and then rises again. Treat the floodlight as a single point and the ground as a curve in the vertical plane through the pole.

Part of the ground is not lit because the hill blocks the light. What is the horizontal extent, in metres, of this unlit part of the ground?

A $\dfrac{3\sqrt{5} - 3}{4}$ **B** 3 **C** 1

D 2 **E** $\dfrac{5}{2}$ **F** $\dfrac{3}{2}$

21. For real numbers p and q, consider the cubic equation

$$x^3 + px + q = 0.$$

Points (p, q) are plotted in a plane with a horizontal p-axis and a vertical q-axis. Let R be the set of all points (p, q) with $-3 \le p \le 0$ for which this equation has three distinct real solutions for x.

What is the area of R?

A $\dfrac{48}{5}$ **B** $\dfrac{12}{5}$ **C** $\dfrac{24}{5}$

D 6 **E** 4 **F** $\dfrac{72\sqrt{3}}{5}$

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22. Nine nonzero real numbers are written in a 3×3 grid. In each row, reading from left to right, the three numbers form an arithmetic progression. In each column, reading from top to bottom, the three numbers form a geometric progression. The number in the top left corner is 2, the number in the centre is 9, and the number in the bottom right corner is 36. What is the sum of all the possible values of the number in the bottom left corner?

A 108

B 90

C 18

D 54

E There are infinitely many possible values.

F 72

23. A point P moves along the line $y = x - 2$. From each position of P, two tangents can be drawn to the parabola $y = x^2$. These two tangents, together with the arc of the parabola between their points of contact, enclose a finite region.

What is the least possible area of this region?

A $\dfrac{2}{3}$ **B** $\dfrac{4\sqrt{2}}{3}$ **C** $\dfrac{7\sqrt{7}}{6}$

D $\dfrac{7\sqrt{7}}{24}$ **E** $\dfrac{7\sqrt{7}}{12}$ **F** $\dfrac{7\sqrt{7}}{4}$

24. A piano tuner experiments with a scale built from perfect fifths. Raising a note by a perfect fifth multiplies its frequency by $\frac{3}{2}$, and raising it by an octave multiplies its frequency by 2. For a positive integer n, say that a stack of n fifths *nearly closes* if there is an integer m for which

$$2^{-1/40} < \frac{(3/2)^n}{2^m} < 2^{1/40},$$

so that n fifths come within one fortieth of an octave of a whole number of octaves. You may use $\log_2 3 = 1.58496$, correct to 5 decimal places. Which of the following is the complete list of the values of n with $1 \le n \le 60$ for which a stack of n fifths nearly closes?

A 12, 41, 53

B 41, 53

C 12 only

D 12, 53

E 12, 24, 36, 48, 60

F 12, 29, 41, 53

G 5, 12, 17, 41, 53

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25. The curve C has equation $y = 3x - x^3$. It is traced onto a sheet of tracing paper, and the sheet is turned over and put back down so that the traced copy becomes the reflection of C in the line $y = x$. The copy is then the curve with equation

$$x = 3y - y^3.$$

At how many points do the curve C and its reflected copy meet?

A 1 **B** 5 **C** 9

D 6 **E** 3 **F** 7

26. Inversion in the circle $x^2 + y^2 = 16$ is the transformation that sends each point P other than the origin O to the point P' on the ray from O through P for which $OP \times OP' = 16$. Let R be the region consisting of all the points (x, y) with $x > 4$ that lie outside the circle $(x - 5)^2 + y^2 = 1$. The images P' of all the points P of R form a region R'. What is the area of R'?

A 3π

B The area of R' is infinite.

C $\dfrac{32\pi}{9}$

D 4π

E $\dfrac{4\pi}{9}$

F $\dfrac{40\pi}{9}$

G $\dfrac{140\pi}{9}$

27. Alice and Bob play a game. First Alice chooses a real number a with $0 \leq a \leq 5$ and tells Bob. Then Bob chooses a real number b with $0 \leq b \leq 2$, and Bob pays Alice

$$b^3 - 3ab + a^2$$

pounds (if this amount is negative, Alice pays Bob instead). Bob always chooses b to make the amount he pays as small as possible, and Alice, knowing this, chooses a to make that amount as large as possible.

How many pounds does Bob pay Alice?

A 3 **B** 0 **C** 8

D $25 - 10\sqrt{5}$ **E** 25 **F** $-\dfrac{27}{16}$

End of paper

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